

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA43

NAME: Naveen G
REGNO: 192411019

COURSE OUTCOME COVERED

CO-2 : Apply CSS layout and styling techniques to create visually appealing and responsive web interfaces, and use JavaScript to enable interactive client-side behavior. (L3)

Assessment Tool : Short Answer Test

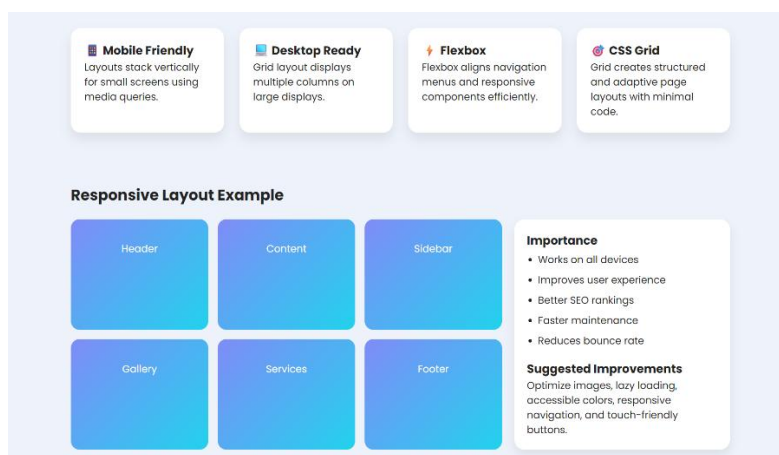
Weightage: 10%

QUESTIONS:

Total Marks: 20

Question 1 (5 Marks)

Suppose that a company wants its website to work seamlessly on mobile, tablet, and desktop devices. Explain the concept of responsive web design and its importance. Describe the key CSS techniques used to achieve responsiveness, and explain how CSS Flexbox or Grid can be used to design adaptive layouts. Illustrate your answer with a simple example layout and suggest improvements to enhance user experience across different devices.



Question 2 (5 Marks)

Suppose that a webpage contains a form where user input must be validated before submission. Explain the concept of client-side validation and its purpose. Describe how JavaScript event handling is used in validation and explain the role of regular expressions in validating inputs such as email or phone numbers. Provide a suitable example and suggest improvements for better usability and security.

Client-Side Validation

Client-side validation checks user input inside the browser before submitting the form. It improves user experience, reduces unnecessary server requests, and provides immediate feedback.

Concepts Covered

- JavaScript Event Handling
- Regular Expressions (Regex)
- Email Validation
- Phone Number Validation
- Password Strength Meter
- Better Usability & Security Tips

Improvements

- Display real-time error messages.
- Use HTTPS for secure communication.
- Validate again on the server.
- Prevent weak passwords.

Registration Form

Name

Email

Phone

Password

☐ Show Password

Question 3 (5 Marks)

Suppose that a webpage layout breaks when the browser window is resized. Explain the CSS box model and its components, and describe how CSS positioning techniques affect layout structure. Identify possible causes of layout breaking and explain how media queries can be used to resolve these issues. Suggest best practices to ensure a stable and responsive layout.

CSS Box Model, Positioning & Responsive Layout

Assessment Question 3 Demonstration

Resize the browser window to observe the responsive layout.

Interactive CSS Box Model

The CSS Box Model consists of **Content**, **Padding**, **Border**, and **Margin**. Using `box-sizing: border-box` helps prevent layout breaking.

Why Layouts Break?

- Fixed width elements
- Large images without max-width
- Missing box-sizing
- Improper margins and padding
- No media queries
- Absolute positioning misuse

Media Queries

Media queries apply different CSS rules depending on screen size, ensuring that layouts adapt smoothly for mobile, tablet, and desktop devices.

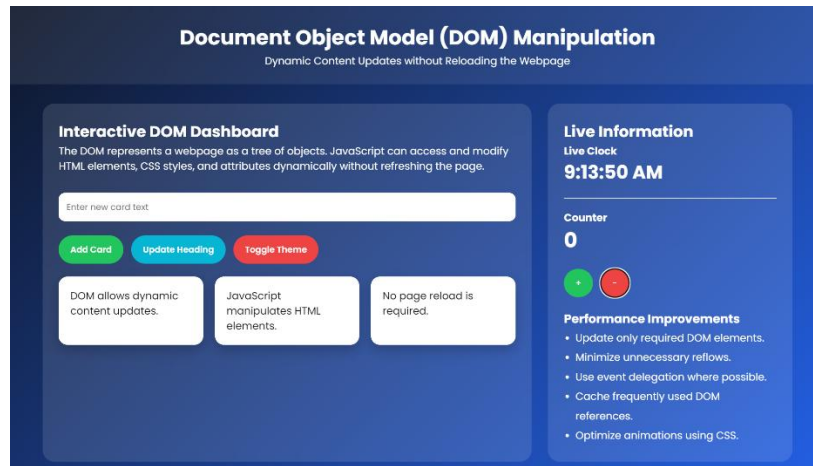
Best Practices

- Use Flexbox and CSS Grid.
- Use responsive units (% , rem , vw).

Desktop (1280px)

Question 4 (5 Marks)

Suppose that a website requires dynamic content updates without reloading the entire page. Explain the concept of the Document Object Model (DOM) and its role in web development. Describe how JavaScript can be used to manipulate the DOM for updating webpage content dynamically. Provide an example and suggest improvements to enhance performance and user experience.



Code of Conduct:

I Naveen G / 192411019 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding

Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand